Immunologically Privileged Sites and Peripheral Tolerance

Activation of autoreactive lymphocytes can be further prevented by the relative paucity of antigen-presenting cells (APCs) in certain organs and the minimal expression of MHC molecules in these tissues. These sites are termed **immunologically privileged** because even tissue grafts do not elicit immune responses when transplanted into them. Immunologically privileged sites possess additional mechanisms that actively suppress immune activation. These include:

- Production of **transforming growth factor-β (TGF-β)**, which suppresses immune responses—such as in the anterior chamber of the eye.
- Constitutive expression of **Fas ligand**, which induces apoptosis of infiltrating activated lymphocytes.

Despite the expression of autoantigens in some tissues and the presence of autoreactive lymphocytes in the circulation, recognition of autoantigens presented by MHC molecules on APCs is typically insufficient to activate these cells. Full T-cell activation requires a **second signal**—co-stimulation—provided by molecules such as **CD80** and **CD86** on APCs binding to **CD28** on T cells. In the absence of this co-stimulatory signal, autoreactive T cells that receive signal one (through the T-cell receptor) undergo **apoptosis** or become **anergic**.

Upregulation of CD80 and CD86 on APCs generally occurs only in response to inflammatory stimuli, such as those triggered by pathogens during infection. Therefore, presentation of self-antigens in the absence of inflammation does not lead to autoreactive T-cell activation. In contrast, pathogen-derived peptides are presented in an inflammatory context, leading to effective immune activation. However, if self-antigens are presented at sites of inflammation, this protective mechanism may be bypassed, potentially triggering autoimmunity. Fortunately, such events are rare, and autoimmunity is usually avoided.

Autoreactive lymphocytes that do become activated in the periphery can be controlled through multiple mechanisms:

- **Deletion via apoptosis**, mediated largely by the **Fas–Fas ligand** pathway on T cells.
- **Inhibition of activation** through inhibitory receptors such as **CTLA-4** and **PD-1**, which are expressed on activated T cells and serve to downregulate immune responses.

The critical role of CTLA-4 and PD-1 in preventing autoimmunity is supported by mouse models in which deletion of these genes results in spontaneous autoimmune disease. Furthermore, genetic polymorphisms in *CTLA-4* and *PDCD1* (encoding PD-1) have been associated with autoimmune disorders such as **systemic lupus erythematosus (SLE)** and **rheumatoid arthritis (RA)**.

Therapeutically, targeting these inhibitory pathways has emerged as a strategy for treating autoimmune diseases. For example, **abatacept**, a recombinant fusion protein consisting of the extracellular domain of human CTLA-4 linked to the Fc portion of human IgG1, is approved for the treatment of RA. Abatacept mimics CTLA-4 function by competing with CD28 for binding to CD80/CD86 on APCs, thereby blocking co-stimulation and selectively inhibiting T-cell activation.

In addition to these mechanisms, the activation of autoreactive lymphocytes is actively suppressed by specialized subsets of T cells known as **regulatory T cells (Tregs)**, discussed below.

Regulatory T Cells

The idea that certain T cells can regulate or suppress immune responses originated in the early 1970s, when Gershon described **suppressor T cells**. However, these cells remained poorly defined, and their mechanisms of action were unclear, leading to skepticism and limited interest for many years.

The concept was revived more than a decade later with key findings showing that **CD4$^+$ T cells expressing CD25** (the α chain of the interleukin-2 receptor) could modulate immune responses. This work, notably by Sakaguchi and others, helped establish the existence of a distinct regulatory population now known as **regulatory T cells (Tregs)**. Additional regulatory subsets were identified based on their production of immunosuppressive cytokines such as **TGF-β** and **IL-10**.

Today, Tregs are recognized as central players in maintaining peripheral tolerance and preventing autoimmune diseases.